

Thesis figures: what to insert, where, and what to replace

Companion to `THESIS_REVISION_PLAN.md`. Generated 30 July 2026 from `thesis1.md` and the full project folder. Every figure listed here is in `Doc images/` with a name beginning `NEW_fig_`. Numbers quoted in the captions were measured by running the code, not estimated.

Read first. Three figures in your draft cannot be generated from data because they need photographs of the physical dome that do not exist in the project folder. See **Section A** below. That is the only gap in the figure set and it needs your camera, not mine.

A. Photographs you must take yourself

Your draft has three figure slots that require photographs of the built dome. Searching the whole project folder turns up axis-test photographs but **no photograph of the assembled dome**. The captures in `captures/` are all through the bare lens.

Draft slot	What it must show	Status
Fig 4.1 "Image of constructed dome"	The assembled eight-gore dome sitting on its octagonal lid mount, photographed from outside in daylight	Missing. Photograph it.
Fig 4.3 "Final Dome Construction Prototype"	The complete rig: dome over the camera, levelled, with the North mark visible on the rim	Missing. Photograph it.
Film axis confirmation	Two sheets crossed, one half bright and one half dark	You already have this. Use <code>Doc images/WhatsApp Image 2026-07-28 at 20.46.01.jpeg</code> (in <code>Other images/</code>), or the outdoor version <code>12.37.23.jpeg</code>

When you photograph the dome, put a ruler or a coin in the frame for scale, and take one shot looking down the optical axis so the eight gores and the apex are visible. Also measure the finished dome radius and the lid circumradius with a ruler and write both numbers into Section 4.2; the confirmed cutting guide has blank lines for exactly these.

B. Complete figure inventory

File in	Goes in	Replaces or fills
<code>NEW_fig_3_1_azimuth_conventions.png</code>	§3.1, end	New. No figure existed. Fixes the thesis's

		most repeated source of confusion.
NEW_fig_3_3_dop_vs_scattering_angle.png	§3.3, by the DoP equation	New. Illustrates the equation that currently stands alone.
NEW_fig_4_1_gore_template_R40.png	§4.2	Replaces Fig 4.2 and the wrong $R = 9$ cm text.
NEW_fig_4_4_processing_pipeline.png	§4.4	Fills the placeholder "[Figure 4.2: processing pipeline]".
NEW_fig_5_1_exposure_bracket.png	§5.1	New. Supports the exposure discussion with measured data.
NEW_fig_6_2_azimuth_profile.png	§6.2, after Fig 6.2	New. Proof for the corrected dark-band claim.
outputs/polarizer_band_demo.png	§6.2 (Fig 6.2)	Re-insert. The version in your draft carries a false caption printed inside the image.
NEW_fig_6_3a_error_by_condition.png	§6.3, before Table 6.1	New. Visual summary of the six-condition matrix.
NEW_fig_6_3b_error_distributions.png	§6.3, after Table 6.1	New. Shows the bimodality that explains the mean-median gap.
NEW_fig_6_3c_error_vs_elevation.png	§6.3, optional	New. Ties accuracy to the high-latitude elevation range.
NEW_fig_6_4a_exposure_disc_tradeoff.png	§6.4 (new section)	New. The central result of the new section.
NEW_fig_6_4b_captured_frames.png	§6.4 (new section)	New. Shows what the captures actually contain.
NEW_fig_7_1_confidence_vs_error.png	§7.1	New. Evidence for the confidence-metric limitation.
NEW_fig_7_2_seasonal_solar_arcs.png	§7.2	New. Supports the October discussion.

C. Section by section through the thesis

Front matter

Four placeholders an examiner sees before any content.

1. **Abstract** is the literal string "Something here, maybe 3 paragraphs."
2. **Keywords** are from a different thesis: "large language models, ChatGPT, computing education, learning analytics, object-oriented programming". Replace with: sky compass, polarised skylight, celestial navigation, GNSS-denied navigation, fisheye camera, bio-inspired robotics, dorsal rim area, high-latitude operation.
3. **AI declaration** still reads "[Tool Name, e.g., Grammarly]" and "[Tool Name, e.g., ChatGPT v4.0]". Name the tools you used.
4. "45 p., 1 app. p." on the title page. Recount once Chapter 8 and the appendix exist.

Chapter 1, Introduction

- delete A duplicated paragraph. Lines 450 to 460 and 462 to 484 both explain the sky compass and the dorsal rim area. Keep the second, which has the sunglasses analogy, and delete the first.
- **typos** "environemental" to "environmental"; "Animals use a variety of changes such as visually sited landmarks" to "cues such as visual landmarks"; "However, Navigation without a satellite fix" lowercase; missing full stop at "from just one direction in the sky This distributed sensing"; "navigated from very small distances to a difference terrain" needs rewriting.
- cite The claim that Finnish platforms suffer GPS degradation during geomagnetic storms is stated flatly with no source. Cite it or soften to "are reported to".

Chapter 2, Literature Review

- fix Line 807: "The table 2.3 shows" should be "Table 2.1 compares". There is no Table 2.3.
- fix Table 2.1, your own row, reads "**Reported in Chapter ??**". Fill it, but note that every other row reports a *field-measured* accuracy while yours is *simulation*. Suggested cell: "Simulation only at time of writing: median absolute azimuth error 0.08 deg clear sky, 0.78 deg thin cloud, 4.10 deg heavy cloud (Table 6.1). Not directly comparable with the field results above."
- **refs** [7] and [8] are the same Wehner 2003 paper. Merge and renumber.
- check [15] and [16] are flagged in your own text as unverified. Confirm against the published versions or drop them.

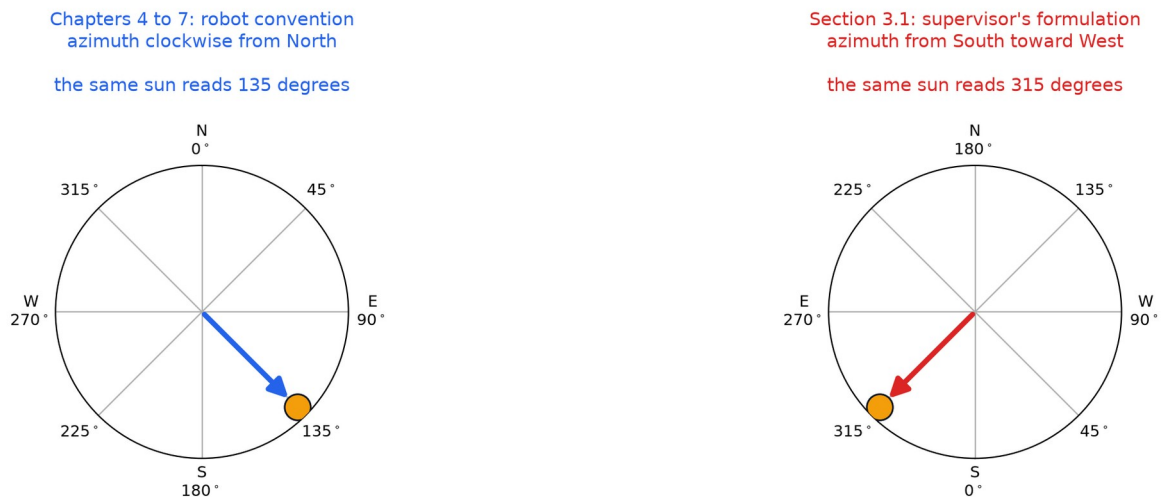
Chapter 3, Theoretical Background

Insert at the end of §3.1, after the paragraph beginning "One point of convention should be recorded here".

Caption: The two azimuth conventions used in this thesis, shown for a single physical sun direction. The derivation

of Section 3.1 measures azimuth from South toward West, while every heading reported from Chapter 4 onward is measured clockwise from North. The two differ by a fixed rotation of 180 degrees.

The two azimuth conventions in use, shown for one physical sun direction. They differ by exactly 180 degrees, with the North referenced azimuth equal to the South referenced azimuth plus 180 reduced into 0 to 360. A mixed convention still runs and still returns a plausible number.

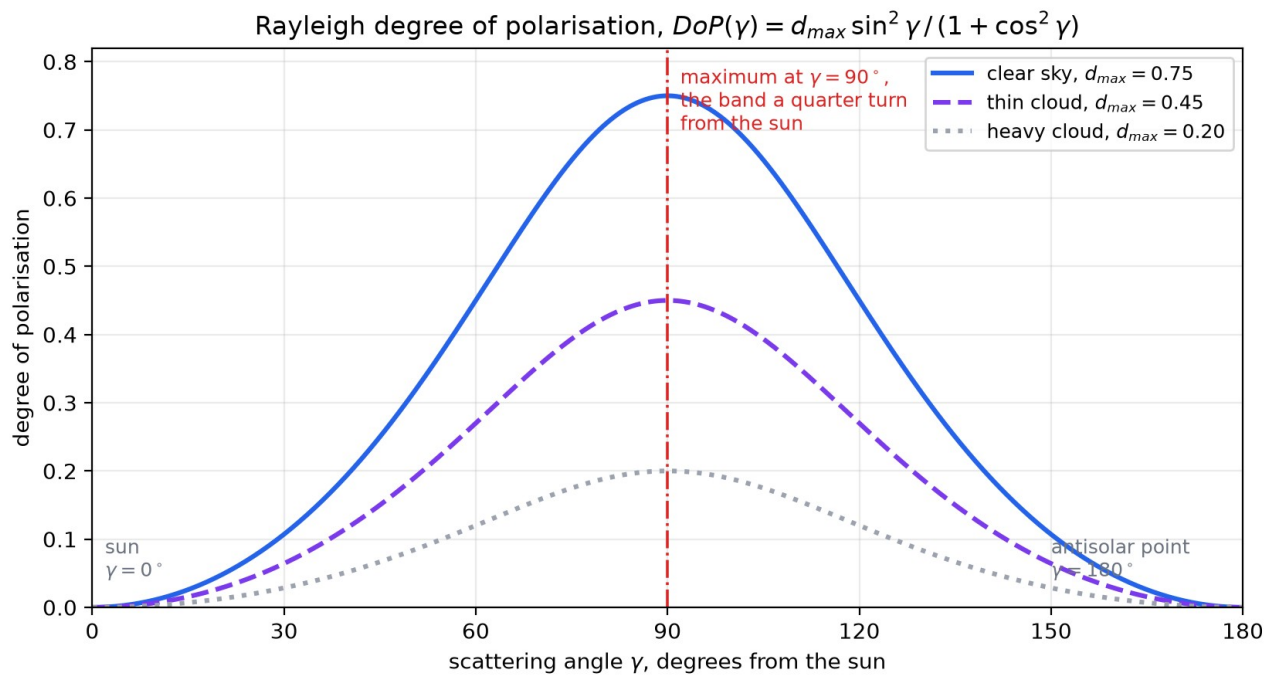


NEW_fig_3_1_azimuth_conventions.png

- fix Line 1169 is garbled: "the sunrise elevation at local noon at the latitude changes by only about 47 degrees". Intended: *the solar elevation at local noon varies by only about 47 degrees over the year, from roughly 6 degrees at the winter solstice to 53 degrees at the summer solstice.*
- fix Line 1114 says the model returns "6.11 and 52.97". Your README and handoff both say 6.110 and 52.986 against analytic 6.108 and 52.988. The Figure 3.1 caption claims agreement within 0.01 degrees, which holds for 52.986 but not for 52.97. Re-run `python3 turku_sun.py` and quote what it prints.
- check The rotation matrix around line 1066 extracts as malformed, with a first row of 1 0 0 and a missing entry. Check it renders correctly in LaTeX.

Insert in §3.3, immediately after the equation for $\text{DoP}(\gamma)$.

Caption: The Rayleigh degree of polarisation as a function of scattering angle, for the three values of the maximum degree of polarisation used in the simulation experiments of Chapter 6. The degree of polarisation vanishes toward the sun and the antisolar point and peaks along the band lying a quarter turn from the sun.



NEW_fig_3_3_dop_vs_scattering_angle.png

Two factual errors in §3.3.

1. **Line 1342 is wrong:** "The iso-AoP contours therefore form concentric rings around the solar position." Rings around the sun are contours of constant *scattering angle*, and therefore of constant *degree* of polarisation. The angle of polarisation is perpendicular to the great circle joining the point to the sun and its contours are not rings. Your own Figure 3.2 caption gets this right, so the text contradicts the figure.
 2. **The dark band claim** at line 1488: the radial dome does *not* put the band on the sun. See §6.2 below for the measurement and the replacement wording.
- refs Line 1441 cites "Gkanias et al. [5]", but [5] is Steinhoff and Schiele on dead reckoning. Should be [13].
 - refs Line 1501 cites "Gkanias et al. [2]", but [2] is Muheim on migratory birds. Should be [10].
 - delete Lines 1340 to 1350 and 1370 to 1378 repeat the same explanation of the e-vector and the rigid rotation. Merge them.

Chapter 4, System Design

Section 4.2 describes a dome you did not build. The draft specifies $R = 9$ cm with gores 14.1 cm long and 7.1 cm

wide, two per sheet across four sheets. That is the superseded dome_cutting_guide.html. The dome you built follows dome_cutting_guide_FINAL_CONFIRMED.pdf.

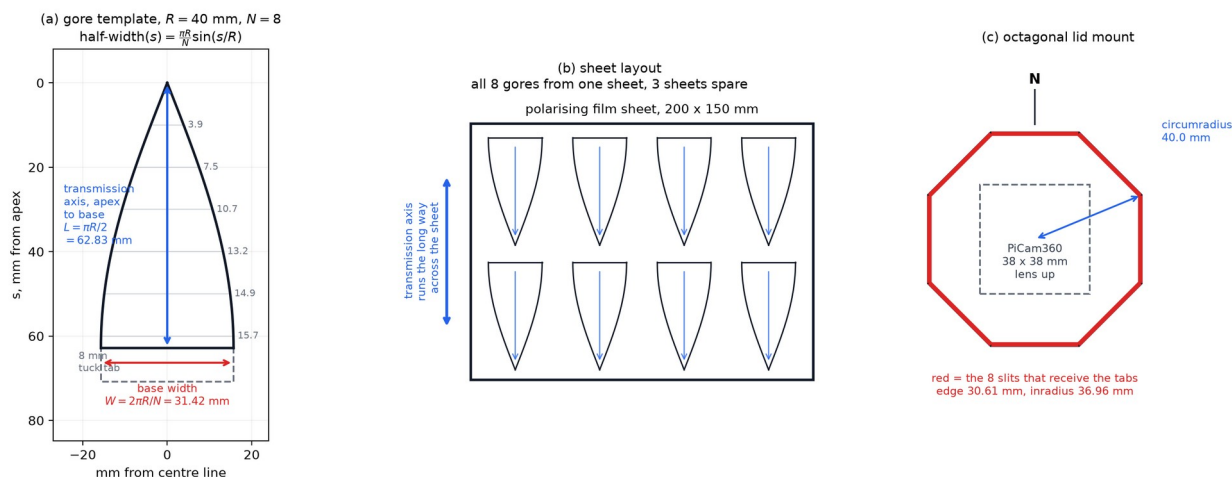
Quantity	Draft says	Built dome
dome radius R	9 cm	40.0 mm
gore length $L = \pi R / 2$	14.1 cm	62.83 mm
gore base width $W = 2 \pi R / N$	7.1 cm	31.42 mm
sheets of film	4, two gores each	1 of 4, all eight gores, three spare
mount	"a flat ring holds the base circle open"	octagonal box-lid, circumradius 40.0 mm, edge 30.61 mm, inradius 36.96 mm, eight slits, 8 x 28 mm tabs

The formulas stay. Every number derived from them changes.

Replaces the existing Figure 4.2 in §4.2.

Caption: Fabrication geometry of the polarising dome. The hemisphere of radius 40 mm is covered by eight identical gores, each of length 62.83 mm and base width 31.42 mm, whose half-width at a distance s from the apex follows $(\pi R / N) \sin(s / R)$. All eight gores are cut from a single 200 by 150 mm sheet with the transmission axis running from apex to base, so that in the assembled dome the axes fan radially outward. The base tabs fold through eight slits cut along the edges of a regular octagon of circumradius 40 mm.

Dome fabrication geometry. Hemisphere of radius $R = 40$ mm covered by $N = 8$ identical gores, cut with the transmission axis running apex to base so the axes fan radially.

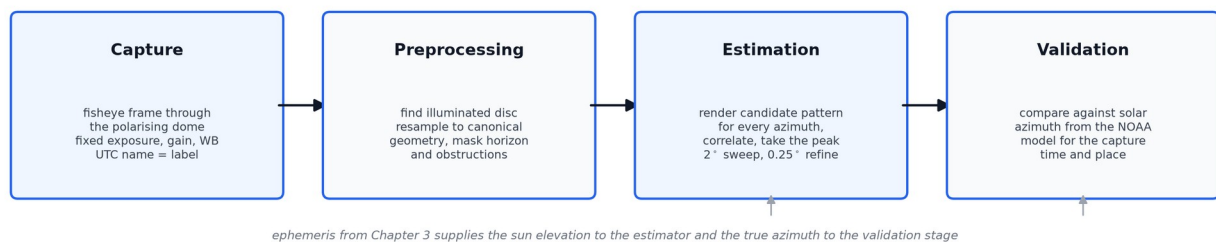


Also rewrite the axis-determination paragraph. The draft says the axis was found "with a simple crossed polarizer test". That test leaves a 90 degree ambiguity between the pass axis and the block axis, and you resolved it with three independent observations: glare off a horizontal surface, an LCD display, and polarising sunglasses with a known axis. All three agreed that the axis runs parallel to the long 200 mm edge. Write that up; it is a much stronger methods paragraph and you have the photographs.

Figure numbering collides three ways in Chapter 4. The caption at line 1611 is "Figure 4.1: Image of constructed dome", an inline placeholder at 1619 calls the gore template "Figure 4.1", the caption at 1647 calls it "Figure 4.2", and §4.4 calls the pipeline "Figure 4.2" as well. Proposed clean set: **4.1** gore template (above), **4.2** film axis confirmation photograph, **4.3** assembled dome on its mount, **4.4** mounted rig outdoors, **4.5** processing pipeline.

Fills the placeholder in §4.4, "[Figure 4.2: processing pipeline]".

Caption: The processing pipeline. A frame is captured through the dome at fixed exposure and white balance and named with its coordinated universal time, the sky region is isolated and each retained pixel associated with a sky direction, the observed pattern is compared against a family of rendered patterns to yield an azimuth and a confidence, and the result is validated against the solar azimuth computed for the time and place of capture.



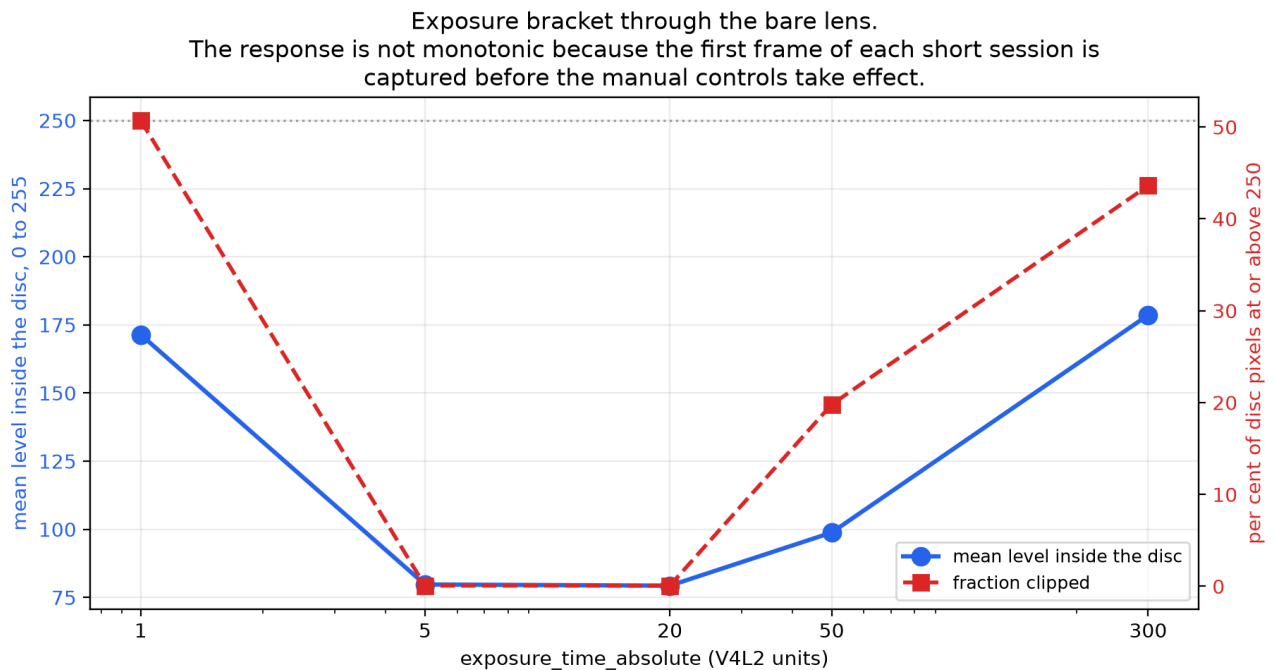
- **typos** §4.3: "PiCam 360" to "PiCam360"; "computer vison" to "computer vision".
- honesty §4.3 describes a mount you do not have. It is a shipping box lid with slits, levelled with a phone level. Say so; the photograph will show it anyway.

Chapter 5, Methods

§5.1, fill the NOTE with the values from captures/*/session.json: device /dev/video4, 1600 x 1200, YUYV, gain 0, white_balance_temperature 5000 K, gamma 100, automatic exposure and white balance disabled, a fixed number of initial frames discarded, and exposure_time_absolute bracketed over 1, 5, 20, 50, 100 and 300. State which value you settled on.

Insert in §5.1 with the exposure discussion.

Caption: Exposure bracket through the bare fisheye lens, measured inside the fitted disc. The response is not monotonic, because the first frame of each short session is captured before the manual controls take effect, which is the settling behaviour described in the text.



NEW_fig_5_1_exposure_bracket.png

Two quantitative claims in §5.1 that do not reproduce.

- Lines 1854 to 1858 state the sky region of Figure 5.1 has "an average intensity of 220 and about 1/3 of the sky pixels being 250 or higher". Measuring inside the fitted disc of that frame gives a mean of **158.5** and **27.6 per cent** at 250 or above. The one-third figure is close; the mean is not. State the exact region you measured or re-measure.
- Line 1900: "the difference in intensity between frames taken 3 seconds apart was about 2.6". 2.6 what? Say "a factor of about 2.6", and quote the two means. The clearest instance in your data is the 26 July pair, where the mean fell from 152.0 to 68.3 across two seconds, a factor of 2.2.
- retitle §5.2 is called "Sun detection and pixel to azimuth mapping" and opens "The sun is not found by looking for its disk." Retitle to "**Pixel to sky direction mapping and heading convention**".
- add §5.3 never defines the confidence metric, yet Chapter 7 criticises it. Define it: the height of the correlation peak above the mean of the correlation curve, in units of the standard deviation of that curve. Also state the search resolution: a 2 degree sweep over the full circle, then a 0.25 degree refinement

within plus or minus 2 degrees.

- move §5.3 promises "A learned regressor trained on synthetic images is also considered". There are no results for it. Move the sentence to §7.3.
- add Move the "true by construction" caveat from Chapter 7 into §5.3, where the reader first meets the estimator.

§5.4 claims something the code does not do. The section says the generator is driven by sun positions "taken from the real record of the sun over Turku". That is true of `evaluate.py`. It is **not** true of `generate_dataset()`, which samples azimuth uniformly over 0 to 360 degrees. Running it confirms this: labels span 0.1 to 359.8 degrees, including roughly 100 degrees of azimuth the sun never occupies at Turku.

Two corrected datasets have been generated for you: `outputs/dataset_turku_flat/` and `outputs/dataset_turku_radial/`, each 2000 frames drawn from the 5972 real daylight positions, spanning azimuth **50.2 to 310.7 degrees** and elevation 5.0 to 53.0 degrees, with the UTC timestamp kept as a label column. Use these and delete `outputs/dataset_flat/`.

Chapter 6, Results

§6.1 mixes two datasets in one passage. It quotes azimuth "35 to 325 degrees" and "5972 daylight solar positions" together. These are different populations:

Set	Count	Azimuth range
elevation ≥ 0 deg, plotted in Figure 6.1	6740	35 to 325 deg
elevation ≥ 5 deg, used by the estimator	5972	50 to 311 deg

State both explicitly and explain the 5 degree floor. Separately, the window 14 May to 14 July is called "the two months leading up to the capture campaign", but the campaign ran 26 to 30 July. Reword, or shift to 31 May to 31 July, which gives $n = 5965$ with identical ranges.

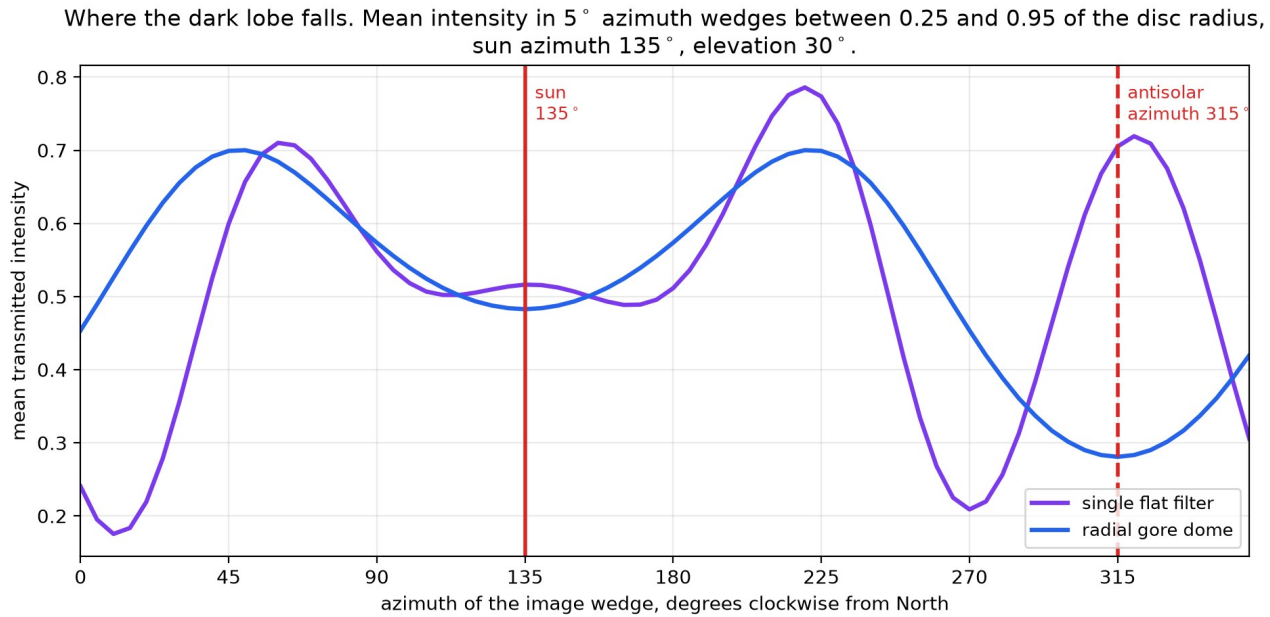
§6.2 asserts something the figure contradicts. The draft says the radial dome "places the band so that it points toward the sun", in §3.3, §4.1 and §6.2. Measurement: binning the rendered image into azimuth wedges puts the darkest wedge at the **antisolar** azimuth every time (sun 0 to dark 180, 90 to 270, 135 to 315, 225 to 45), across elevations of 10, 30 and 50 degrees. Mean intensity toward the sun is 0.489 against 0.293 toward the antisolar point.

The correct statement: the radial arrangement aligns the dark axis with the solar meridian, because there the sky vector is tangential while the analyser axis is radial. Both ends darken, and the antisolar end darkens more because the degree of polarisation is low near the sun. The brightness asymmetry between the two ends is what identifies which end holds the sun, and it is why the ambiguity returns once cloud flattens the pattern.

Insert in §6.2, after the existing Figure 6.2.

Caption: Mean transmitted intensity in five degree azimuth wedges between 0.25 and 0.95 of the disc radius, for a sun at azimuth 135 degrees and elevation 30 degrees. The flat filter produces two minima fixed by the film axis and unrelated to the sun. The radial dome produces a single deep minimum at the antisolar azimuth of 315 degrees together

with a shallower one toward the sun, so the dark axis marks the solar meridian while the asymmetry between its ends resolves which end holds the sun.



NEW_fig_6_2_azimuth_profile.png

Also re-insert outputs/polarizer_band_demo.png. The version in your current draft has the false claim printed *inside* the image, where an examiner reads it straight off the figure. The regenerated file corrects the caption and adds the solar meridian as a cyan diameter, solid toward the sun and dashed toward the antisolar azimuth, so the claim is visible rather than merely asserted. Note in the caption that the antisolar *point* lies below the horizon and is not in the image; what is marked is the antisolar *azimuth* at the disc edge.

\$6.3, replace Table 6.1 entirely. The draft has four of six cells at $n = 60$. The full matrix has been run at $n = 200$:

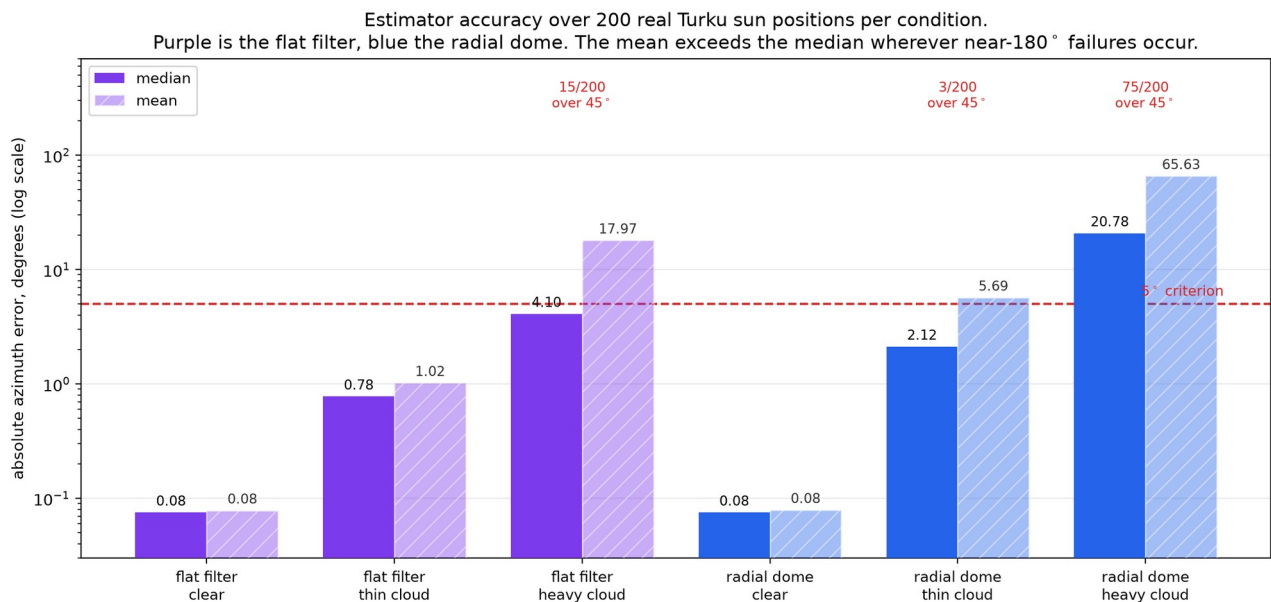
Condition	Optic	Mean	Median	P90	Max	Within 5 deg	Conf.	Gross
clear	flat	0.078	0.075	0.138	0.272	100.0 %	1.80	0
thin cloud	flat	1.020	0.777	2.188	4.224	100.0 %	1.80	0
heavy cloud	flat	17.972	4.103	15.639	179.951	56.5 %	1.75	15/200
clear	radia	0.079	0.075	0.142	0.269	100.0 %	1.78	0

thin cloud	radia l	5.687	2.120	8.432	179.91 1	75.0 %	1.74	3/200
heavy cloud	radia l	65.63 4	20.783	177.32 4	179.83 9	18.5 %	1.53	75/200

Report the median as the primary statistic. At $n = 60$ the radial thin cloud mean was 11.59 degrees; at $n = 200$ it is 5.69, while the median held at 2.12 in both. The mean is hostage to a handful of near-180 degree failures and is not stable at small n . This also explains the rows where the 90th percentile is smaller than the mean.

Insert in §6.3, before Table 6.1.

Caption: Absolute azimuth error over 200 sun positions drawn from the real Turku daylight record, for the two optical arrangements under three sky conditions. Bars show the median and the mean on a logarithmic scale. The mean exceeds the median wherever near-180 degree failures occur, and the count of such failures is given above each affected condition.



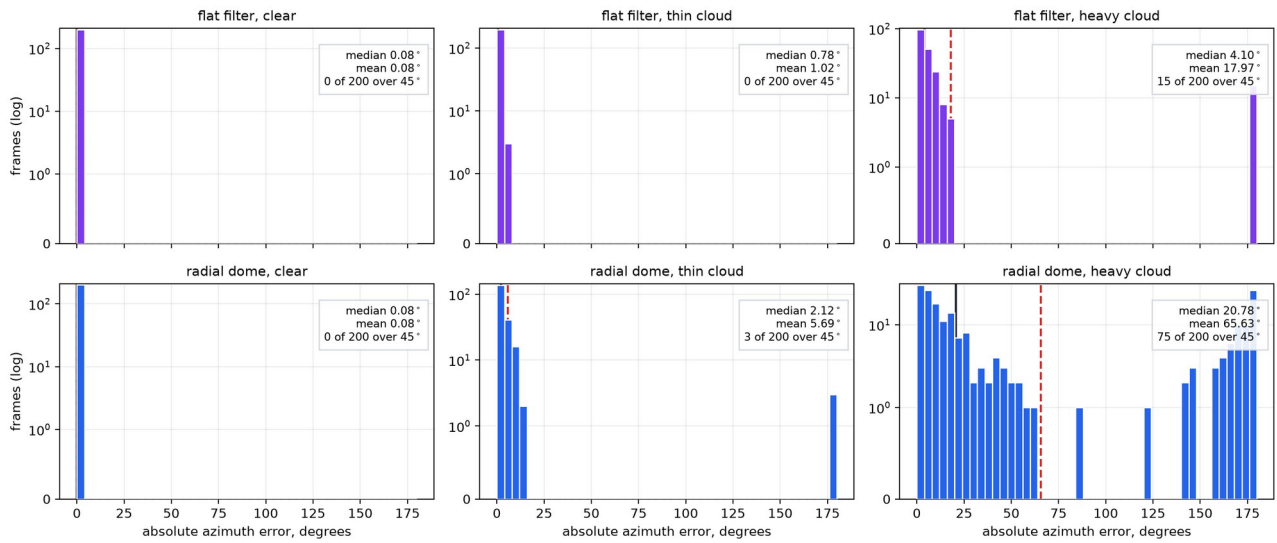
NEW_fig_6_3a_error_by_condition.png

Insert in §6.3, after Table 6.1.

Caption: Distribution of the absolute azimuth error in each condition. The vertical scale is logarithmic. Under a clear sky the error is tightly concentrated below a tenth of a degree, while under cloud the distribution becomes bimodal, with most frames accurate and a minority failing by close to 180 degrees, which is the solar and antisolar ambiguity

discussed in Section 7.1.

Error distributions. Solid black line is the median, dashed red the mean. The distribution is bimodal wherever the sky is degraded: most frames are accurate and a minority fail by close to 180 degrees, which is the solar and antisolar ambiguity.

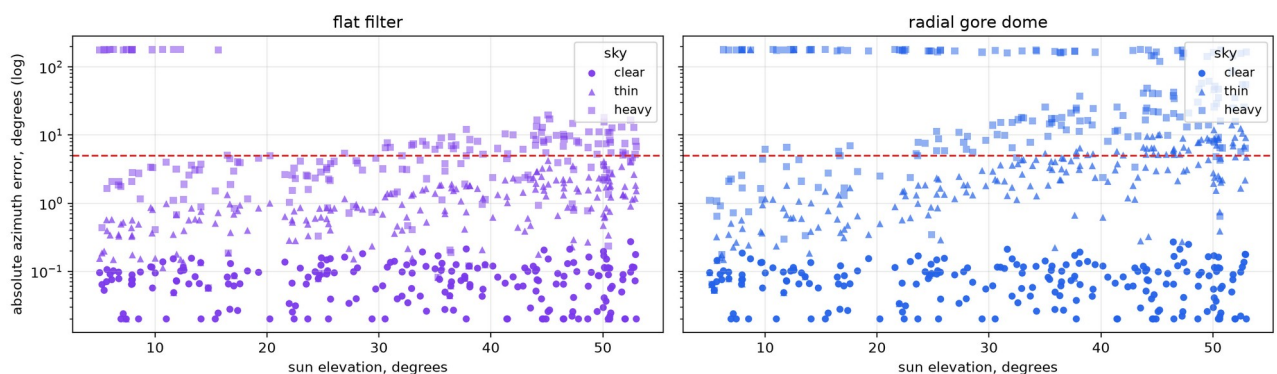


NEW_fig_6_3b_error_distributions.png

Optional, §6.3. Include if you have room; it ties the accuracy to the high-latitude geometry.

Caption: Absolute azimuth error against sun elevation. At Turku the elevation never exceeds 53 degrees, so the estimator is never tested against a sun near the zenith. Failures under cloud are spread across the elevation range rather than concentrated at low sun.

Error against sun elevation. The high-latitude range at Turku is 5 to 53 degrees, so the estimator is never tested on a sun near the zenith. Failures under cloud are spread across the elevation range rather than concentrated at low sun.



The radial dome result must be discussed, not buried. Under a clear sky the two optics are indistinguishable.
 Under cloud the radial dome is consistently worse, with 75 gross failures against 15 for the flat filter under heavy cloud.
 The cause is measurable: the radial pattern's contrast is 0.79 against the flat filter's 1.18, so it drowns in clutter sooner.
 This qualifies the design argument rather than destroying it, and reporting it plainly is far stronger than leaving it out.

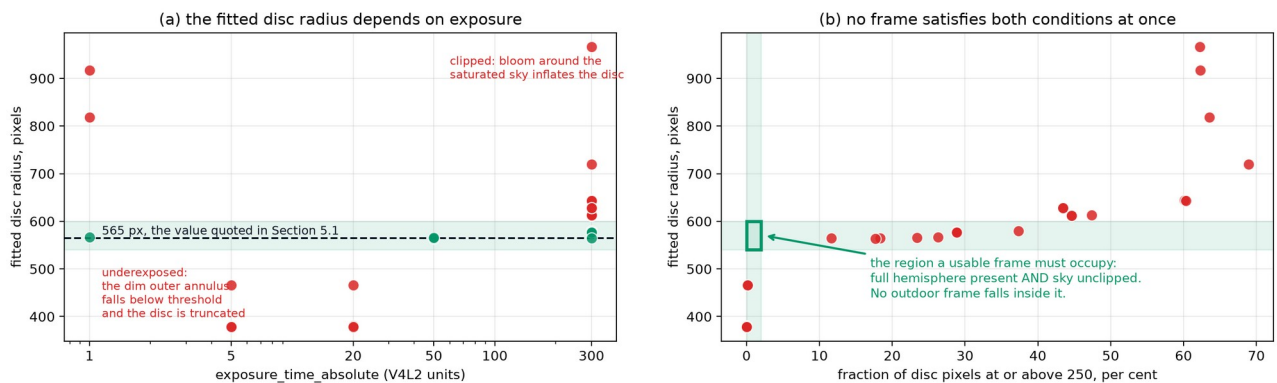
Chapter 6.4, a new section

Chapter 7 refers to "the captured image validation of Section 6.4", but no Section 6.4 exists. A full draft is in SECTION_6_4_DRAFT.md, about three pages, written in the thesis style. Measuring all 32 stored frames established that **none of them is a sky-compass measurement**: the polarising optic is not fitted in any session, the optical axis is not vertical, and the camera was hand held. The section therefore reports what the captures do establish, which is the acquisition chain and the radiometric conditions a real measurement will need.

Insert in §6.4.

Caption: Radiometric limit on the stored captures. Each point is one of the 31 outdoor frames. The fitted disc radius depends strongly on exposure, falling to 378 pixels when underexposure truncates the dim outer annulus and rising to 967 pixels when bloom around a saturated sky inflates it. A usable frame must show the whole hemisphere and leave the sky below saturation, and no captured frame satisfies both conditions.

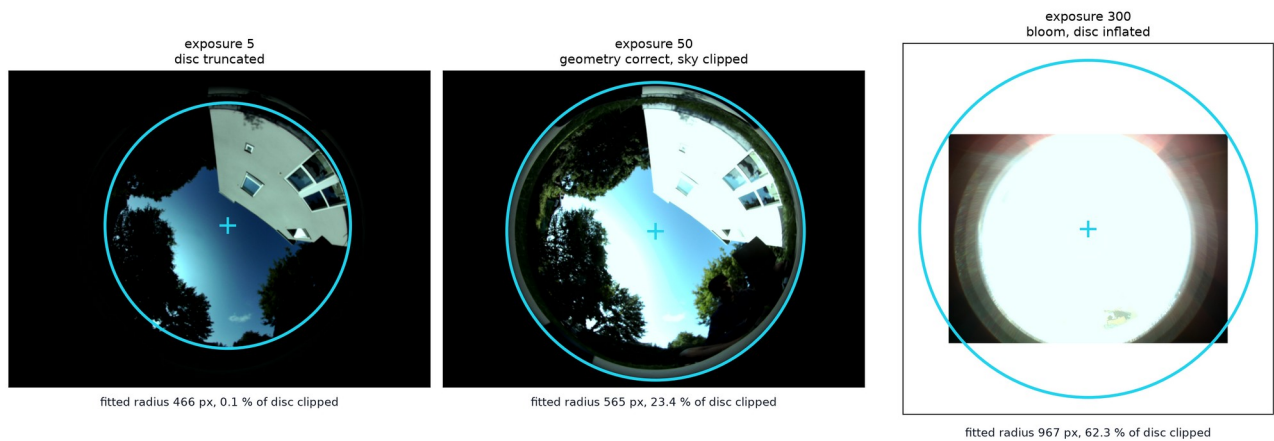
Radiometric limit on the stored captures. Each point is one of the 31 outdoor frames.
 Geometry and exposure cannot presently be satisfied together, which is why no heading can be recovered.



Insert in §6.4.

Caption: Three captured frames with the fitted disc overlaid in cyan, showing the underexposed, correctly exposed and clipped regimes. The polarising optic is not fitted and the optical axis is not vertical, so the ground, a building and the operator occupy much of the field and the usable sky is confined to a wedge.

Representative captured frames with the fitted disc overlaid in cyan. The polarising optic is not fitted and the optical axis is not vertical, so the ground, a building and the operator occupy much of the field and the sky is confined to a wedge.



[NEW_fig_6_4b_captured_frames.png](#)

Confirm two things before submitting. First, that the polarising optic really was absent from every session; this is read off the images, since no session recorded it. Second, whether `image_up_heading_deg_from_north: 0.0` in the 30 July session is a compass reading or the script default. The draft treats it as a default.

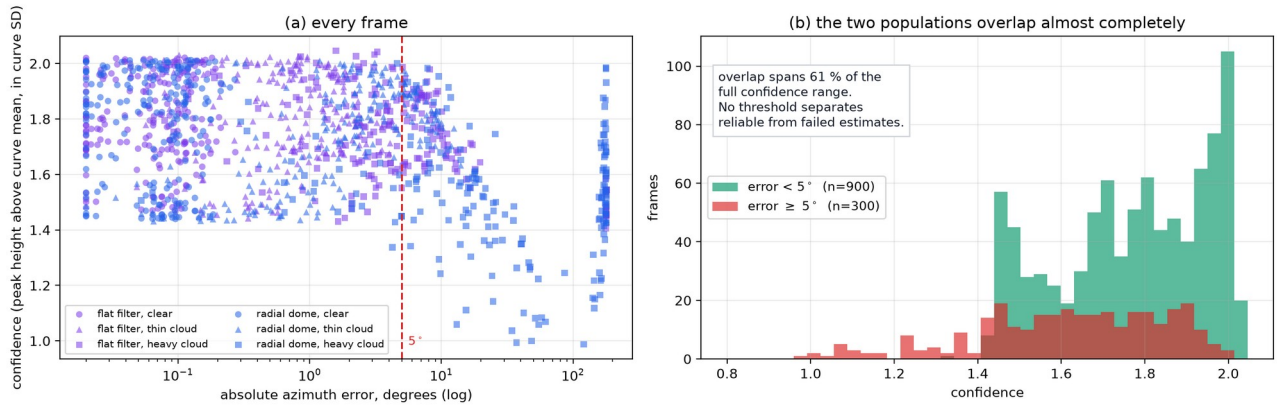
Consequential edit: line 2353 of Chapter 7 says "only the captured image validation of Section 6.4 tests the latter". Reword, since 6.4 no longer validates against the sky.

Chapter 7, Discussion

Insert in §7.1, with the confidence-metric limitation.

Caption: The confidence metric fails to separate reliable estimates from failed ones. Across conditions whose mean error spans nearly three orders of magnitude the mean confidence varies by about fifteen per cent, and the distributions of confidence for accurate and for failed frames overlap across most of the range. Since the purpose of the value was to weight a heading within a fusion filter and to reject solar and antisolar failures, neither is possible until the metric is redefined.

The confidence metric does not discriminate. Mean confidence varies by about 15 per cent across conditions whose error varies by nearly three orders of magnitude, so the value cannot be used to weight a heading in a fusion filter or to reject failures.



NEW_fig_7_1_confidence_vs_error.png

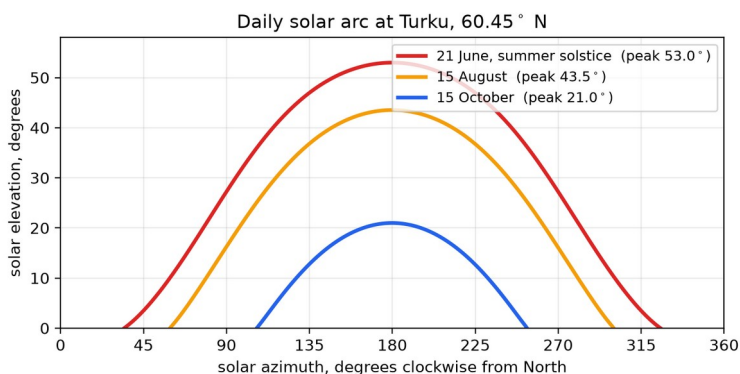
Numbers for the §7.1 text. Mean confidence by condition: 1.80, 1.80, 1.75, 1.78, 1.74, 1.53, against mean errors from 0.078 to 65.6 degrees. Per frame, estimates within 5 degrees have confidence 1.773 plus or minus 0.179 and those beyond 5 degrees have 1.620 plus or minus 0.227, so the two populations overlap across 61 per cent of the range.

Route forward to offer: the ratio of the primary peak to the second highest local maximum would measure the solar and antisolar ambiguity directly, which is the failure that actually occurs, whereas peak height above the mean does not.

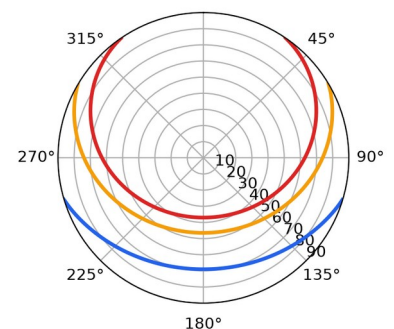
Insert in §7.2, with the October discussion.

Caption: Seasonal change in the sun geometry at Turku. In October the sun rises later and further south, reaches a noon elevation near 20 degrees rather than 53, and sweeps a much narrower range of azimuths, which moves the polarisation pattern toward the horizon where the analysis mask discards it. Repeating the experiments then would test the method against a second and quite different sun geometry at the same site.

Seasonal change in the sun geometry the compass must read. In October the sun rises later and further south, peaks near 20 degrees rather than 53, and sweeps a much narrower azimuth range, which pushes the polarisation pattern toward the horizon.



The same arcs on the sky hemisphere
radius is zenith angle, centre is overhead
0°



Three more wrong citations in §7.3. Line 2467 "Gkanias et al. [5]" should be **[13]**; line 2473 "Stürzl [3]" should be **[11]**; line 2479 "Gkanias and Webb [6]" should be **[14]**. As written they point to Steinhoff, to Sobel's *Longitude*, and to Thrun.

§7.3, expand the rain note. Water on the film and droplets in the air both scatter light through many events, and multiple scattering destroys the linear polarisation that single Rayleigh scattering produces, so rain removes the quantity being measured rather than merely dimming it.

Chapter 8 and the appendix

Chapter 8, Conclusion, is empty. About one page. Restate the question, state what was built and validated, give the headline numbers, state plainly what was not achieved (calibrated outdoor validation, a working confidence metric, the radial dome's expected robustness advantage), and close on what the work leaves for whoever continues it.

Appendix A, "Full Weekly Results", is empty. Either fill it from `outputs/n200_*.txt` or remove it from the contents and from "1 app. p." on the title page.

References

Ref	Problem	Fix
[7], [8]	Identical paper, Wehner 2003, listed twice	Merge and renumber
[11]	"W. Sturzl"	"W. Stürzl"; check the venue
[15], [16]	In preparation, flagged as unverified in your own text	Confirm or drop
[18]	Author given as the GitHub handle "evgkanias"	"E. Gkanias"
[31]	Bare URL, no title or author	Give it a title

D. Order of work, given the deadline

#	Task	Time	Why now
1	Front matter: abstract, keywords, AI declaration	40 min	Read first, currently placeholders
2	Photograph the dome and the mounted rig; measure the radius	20 min	Only you can do this; blocks Figs 4.1 and 4.3

3	§4.2 rewrite with the $R = 40$ mm numbers, drop in the gore figure	45 min	Most serious factual error
4	Fix the five wrong citations, merge [7]/[8], "table 2.3", "Chapter ??"	25 min	Cheap and highly visible
5	Correct the band claim in §3.3, §4.1, §6.2; insert the azimuth-profile figure and the regenerated band demo	40 min	The draft contradicts its own figure
6	Replace Table 6.1; insert the two §6.3 figures; write the prose	40 min	Compute is already done
7	Fill the §5.1 and §5.4 NOTE blocks; insert the exposure figure	45 min	Removes every remaining placeholder
8	Paste in §6.4 with its two figures	30 min	Stops Chapter 6 being simulation only
9	Chapter 8 conclusion	60 min	Required
10	§7.1 and §7.2 figures and text; renumber Chapter 4 figures; appendix decision	45 min	Polish

Items 1 to 5 change an examiner's impression most per minute spent. Item 9 is not optional. Items 10 and the optional Figure 6.3c are the first things to drop if time runs short.